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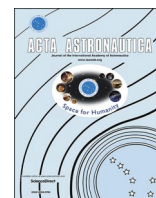
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Review of in-situ oxygen extraction from lunar regolith with focus on solar thermal and laser vacuum pyrolysis

Jack Robinot^a, Sylvain Rodat^a, Stéphane Abanades^{a,*}, Alexis Paillet^b, Aidan Cowley^c

^a CNRS, Processes, Materials and Solar Energy Laboratory (PROMES-CNRS), 7 Rue du Four Solaire, Font Romeu, 66120, France

^b National Centre for Space Studies (CNES), 18 Avenue Edouard Belin, 31401, Toulouse, France

^c European Space Agency, European Astronaut Centre (ESA-EAC), Linder Höhe, Cologne, 51147, Germany

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ABSTRACT

In Situ Resource Utilization (ISRU) is crucial for lowering the cost of space exploration and sustaining human activity on the Moon. Multiple techniques have been studied to extract oxygen from lunar regolith, including electrochemical processes such as molten regolith electrolysis and molten salt electrolysis, as well as pyrometallurgical processes such as hydrogen and carbothermal reduction. However, one attractive technique remains underexplored, namely vacuum pyrolysis. This method utilizes high temperatures and benefits from local vacuum conditions to produce oxygen from lunar regolith composed of various minerals. Vacuum pyrolysis is particularly attractive as it is based on local lunar resources, it requires no external consumables, and it is capable to operate with any lunar soil feedstock composition. Among the possible sources of high-temperature energy, laser and concentrated solar energy are suitable options to drive the endothermal pyrolysis process on the Moon. In particular, using widely available solar energy as process heat source on the lunar surface eliminates the need for intermediate electrical production for the production of oxygen and various metallic compounds from lunar minerals. This review article compiles and examines the various works carried out on this ISRU process, and further provides different aspects to be considered for the development of an integrated and self-sufficient oxygen production process for future applications on the Moon.

1. Introduction

Humanity has reignited a strong interest in space exploration, with the first stage being to step back on the Moon to develop and validate novel technologies. Through its Artemis program, NASA with the collaboration of ESA plans to return to the Moon in 2026 and to establish a human presence by 2030, as a test ground for future Mars exploration. Simultaneously, the CNSA (Chinese national space agency) and Roscosmos are also ambitious to settle on the Moon for space exploration with their International Lunar Research Station project (ILRS) by the year 2030, even though operation plan details remain not communicated. The principal barrier to develop these activities is the huge cost of launching infrastructures and sustaining activities on site. According to Astrobotic [1], shipping 1 kg of payload to the Moon's surface could be as expensive as \$1.2 million. In response, In Situ Resource Utilization (ISRU) has emerged to alleviate these costs. ISRU applications primarily focus on the production of fluids for life support systems, such as oxygen and water, or propellants like oxygen and hydrogen. Additional

applications may involve metal extraction or the use of regolith as a construction material. To date, oxygen extraction from lunar regolith is one of the most extensively studied ISRU processes, underscored by the critical importance of the resource. Lunar-sourced oxygen has numerous uses, with initial emphasis on creating a breathable atmosphere within the base and offsetting losses during operations. Losses may arise from exhaled carbon dioxide, airlock usage, extravehicular activities, and potential leaks. Additionally, oxygen supply could support other ISRU processes such as material processing. A more ambitious use of oxygen is as a propellant, particularly as a rocket engine oxidizer enabling spacecraft to be refueled in space, avoiding the effects of Earth's gravity and dense atmosphere.

This review article aims to provide a detailed outlook of the different studies that have been conducted on the vacuum pyrolysis of lunar regolith and related techniques with its main components for the oxygen production. Firstly, the lunar soil composition will be briefly described; next the main studied processes to produce oxygen from regolith including electrolytic and pyrometallurgical processes will be described

* Corresponding author.

E-mail address: stephane.abanades@promes.cnrs.fr (S. Abanades).

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to provide insights into the interest of developing an underexplored pathway, namely vacuum pyrolysis. Lastly, the different works carried out on the solar and laser vacuum pyrolysis processes will be thoroughly reviewed. Different aspects to be considered for the development of an integrated and self-sufficient lunar process for future applications on the Moon are also provided.

2. Lunar soil

The surface of the Moon is predominantly covered by regolith, a layer of fragmented material produced by meteorite impacts and solar wind bombardment. Lunar minerals are composed of metallic oxides [2]. Almost half of the regolith mass is composed of SiO_2 , with other predominant oxides including Al_2O_3 , FeO , MgO , and CaO . Detailed oxide compositions of regolith from Apollo, Luna, and Chang'e sample missions can be found in Table 1, according to the data of McKay et al. [3], Ling et al. [4], and Chen et al. [5]. Oxygen accounts for approximately 42 % of the mass of the regolith, making it the most abundant element on the lunar surface. Understanding the mineralogical and compositional characteristics of the lunar regolith is crucial for choosing the appropriate process based on the location on the Moon and the local composition of the soil. The Moon's surface is divided into two distinct geological regions: maria and highlands. Maria regions, which are the darker plains primarily located on the near side of the Moon, consist of solidified lava oceans. These regions were formed by the freezing of lava and consist of basaltic plains. Maria regions are sub-classified according to their titanium content. When the soil contains more than 6 % (weight) titanium, it is considered to be high-Ti maria regolith, rich in titanium-bearing minerals such as ilmenite (FeTiO_3). Low-Ti maria are defined by a Ti concentration of less than 6 % (weight), although they still contain higher levels of TiO compared to highlands [6]. Maria regions contain significant amounts of iron, magnesium, and silicates such as pyroxene $[(\text{Ca},\text{Mg},\text{Fe})\text{SiO}_3]$ and olivine $[(\text{Mg},\text{Fe})_2\text{SiO}_4]$. In contrast, highland regions are characterized by a light grey color and higher elevation. These areas were not covered by lava seas and are predominantly composed of plagioclase feldspar minerals like anorthite ($\text{CaAl}_2\text{Si}_2\text{O}_8$). Highlands contain lower concentrations of iron and magnesium compared to maria regions.

3. Oxygen extraction processes utilizing lunar regolith resources

The first studies to utilize lunar resources date back to before the first landing on the Moon. Around 20 techniques have been proposed, and these methods have already been extensively reviewed and compiled by various authors [7–11]. A comparison and an evaluation of the various oxygen production methods can be found in a recent communication [12]. This review article does not aim to provide detailed insights into all these techniques. Instead, they are briefly described to provide context for the underexplored vacuum pyrolysis, which is a specific technique thoroughly reviewed in this work. The state-of-the-art

Table 1
Oxide composition in weight percentage of samples taken on the Moon during Apollo, Luna and Chang'e mission [3–5].

Oxide	Min	Max	Average
SiO_2	40.60	48.10	44.59
TiO_2	0.47	8.40	2.58
Al_2O_3	9.70	28.00	17.82
Cr_2O_3	0.11	0.54	0.32
FeO	4.70	22.80	12.28
MnO	0.00	0.41	0.20
MgO	5.60	13.00	9.16
CaO	10.30	15.80	12.51
Na_2O	0.30	0.70	0.43
K_2O	0.04	0.55	0.18
P_2O_3	0.05	0.51	0.16

conventional methods for oxygen extraction from lunar regolith are detailed, including electrolytic and pyrometallurgical processes, before focusing on vacuum pyrolysis using either laser heating or concentrated solar energy as the high-temperature process energy source.

3.1. Electrolysis processes

By applying an electric current, regolith electrolysis permits the extraction of oxygen and metals. Two major techniques were proposed, including molten regolith electrolysis and molten salt electrolysis.

3.1.1. Molten regolith electrolysis

This technique involves the heating of lunar soil to high temperatures of 1600–2000 °C to create a bath of molten regolith and applying an electric current to the material. Such a process causes the oxides to split. At the cathode, metal cations are reduced to liquid or solid metals, while oxygen anions cross the molten bath and are oxidized at the anode, releasing gaseous oxygen. The electrochemical reactions can be described by the following equations:



Molten regolith electrolysis is a simple process that permits a high oxygen yield of 35 % [13]. This yield is not limited by the composition of the regolith, as all the oxides can be reduced. The process can also separate metals from the regolith by controlling the potential, allowing for the recovery of iron and silicon. Regolith heating could be ensured by Joule effect [14]. However, material selection for the process is crucial, as molten regolith is highly corrosive. To avoid corrosion, un-melted regolith could be used to shield the reactor walls. Regarding electrodes, noble metals and ceramics have been proposed due to their high stability; however, they still require periodic replacement.

3.1.2. Molten salt electrolysis

This reaction involves the electrochemical reduction of lunar regolith, by using a molten salt medium, permitting to operate at lower temperatures than molten regolith electrolysis (the operating temperature is between 800 °C and 1200 °C). It is based on the high ionic conductivity of the salts to facilitate the electrolysis of the oxides and to permit the separation of the metals and the production of oxygen. Molten salt electrolysis originates from the FFC-Cambridge (Fray, Farthing and Chen) [15] used for the production of titanium but in general, this process uses graphite electrodes, which are not suitable. Oxygen indeed reacts with carbon contained in graphite to form CO, which causes two issues: oxygen has to be extracted from CO and electrodes are getting consumed by the reaction. Feasibility of using molten salt electrolysis with SnO_2 electrodes was demonstrated by extracting 96 % of oxygen from lunar regolith simulant while producing metal alloys, and one third of the oxygen was detected as off-gas while the rest was lost due to corrosion of the reactor vessels [16]. Salt selection is also important, it should promote high solubility for the oxides, along with a high mobility for the ions and low solubility for the metals. Reactivity with the anodes and the reactor should be avoided. The salt should have a low melting point, viscosity, and vapor pressure [17]. Different salts were tested including CaCl_2 , LiCl [18], or fluoride [19]. However, the process requires regular replacements of the electrodes when oxidized, and the addition of salt due to their evaporation. Both of them would have to be shipped from Earth.

3.2. Pyrometallurgical processes

3.2.1. Thermodynamics

Pyrometallurgical processes involve oxides reduction to separate oxygen from metals at high temperatures. The thermal reduction reaction can be described by the following equation:



Thermal reduction can be conducted with or without the addition of a reducing agent. Common reducing agents include hydrogen and carbonaceous materials [20–24]. The use of a reducing agent lowers the reaction temperature but also may complexify the process. Reduction reactions are driven by Gibbs free energy change according to the following equations:

$$\Delta G = \Delta H - T\Delta S$$

(4)

$$\Delta H = \Delta U + p\Delta V$$

(5)

Gibbs free energy is influenced by both pressure and temperature. These equations allow to determine the thermodynamic equilibrium of a reaction, indicating whether it is in favor of the reactants or the products. Oxide reduction can only occur spontaneously if ΔG is negative,

however under normal conditions, most metals are found in their oxidized state because the reactions have low Gibbs free energy changes, resulting in equilibrium favoring the oxidized form. The lower the Gibbs free energy, the more stable the oxide. Ellingham [25], based on the approximation that reaction enthalpy and entropy are not depending on temperature, compiled the Gibbs free energy changes of different oxidation reactions as a function of temperature (linear plots) (Fig. 1). This tool allows metallurgists to determine the temperature at which an oxide can be reduced to its corresponding metal.

Reduction of a species by another can only occur if the reducing agent has a lower Gibbs free energy change than the oxide. On the diagram, this is indicated by the oxide curve being above the reducer curve. The best reducer is carbon (C) because it reduces almost all metal oxides by forming CO if the temperature is sufficiently high. At high enough temperatures, the Gibbs free energy of the reduction reaction can become negative, resulting in the spontaneous reduction of the

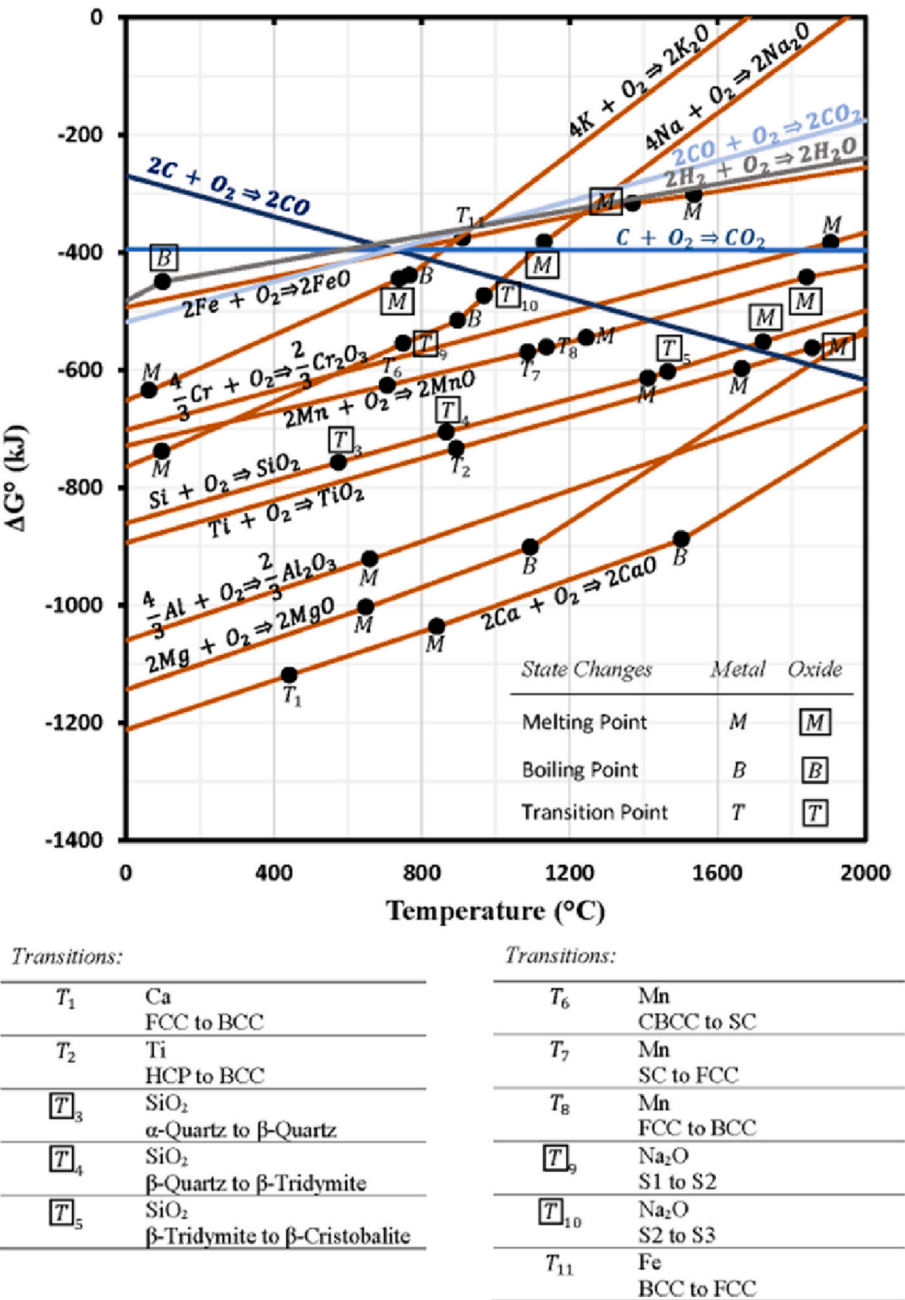


Fig. 1. Ellingham diagram at ambient pressure [26].

oxide, via splitting of the metal and oxygen. At ambient pressure, this reaction would necessitate temperatures above 4000 °C for most of the oxides, with exceptions being K₂O and Na₂O, which dissociate at temperatures between 1600 °C and 2000 °C. Typically, Ellingham diagrams are plotted at 1 atm because most terrestrial processes occur at atmospheric pressure. However, due to the absence of an atmosphere on the Moon, reduction processes could be envisaged to operate at much lower pressures. To be used in lunar conditions, Shaw [26] computed three different Ellingham diagrams at 3×10^{-5} bar, 10^{-10} bar, and 3×10^{-15} bar, based on the Gibbs free energy change formula.

3×10^{-15} bar, based on the Gibbs free energy change formula. Fig. 2 corresponds to the Ellingham diagram calculated at 3×10^{-15} bar, which matches the pressure at the lunar surface.

Lowering the pressure allows the oxides to be reduced at lower temperatures. Using a carbonaceous reducer such as C permits the oxides reduction below 1000 °C at lunar pressure. Below 400 °C, alkaline oxides dissociate into their metallic counterparts, representing a four-fold temperature decrease compared to ambient pressure. Above 1600 °C, the equilibrium favors the metal over the oxide for every

species presented in the diagram. Process yield optimization relies on adjusting the reactor temperature, pressure and reducer.

3.2.2. Carbothermal reduction

The carbothermal reduction involves using a carbon-containing reducer with the regolith to reduce the oxides and extract oxygen. Different reducers, such as solid C, CH₄, and CO, have been considered. However, most efforts have focused on CH₄ reduction due to its significant advantages. This reduction method was proposed as early as the 1960s by Rosenberg [27]. The process requires reacting CH₄ with regolith at high temperatures. At temperatures above 800 °C, CH₄ decomposes into H₂ and solid C upon contact with regolith, with the solid carbon depositing on the regolith surface. The deposited C then reacts with the oxides to form CO and metals, while H₂ remains mostly unreacted. CO and H₂ exit the reactor and are then reacted together with extra H₂ according to the Sabatier reaction, producing CH₄ and H₂O. CH₄ is reinjected back into the carbothermal reactor, and H₂O is electrolyzed to yield O₂ and H₂, which is recycled into the Sabatier reactor. This closed-loop process permits the regeneration of the reductant and

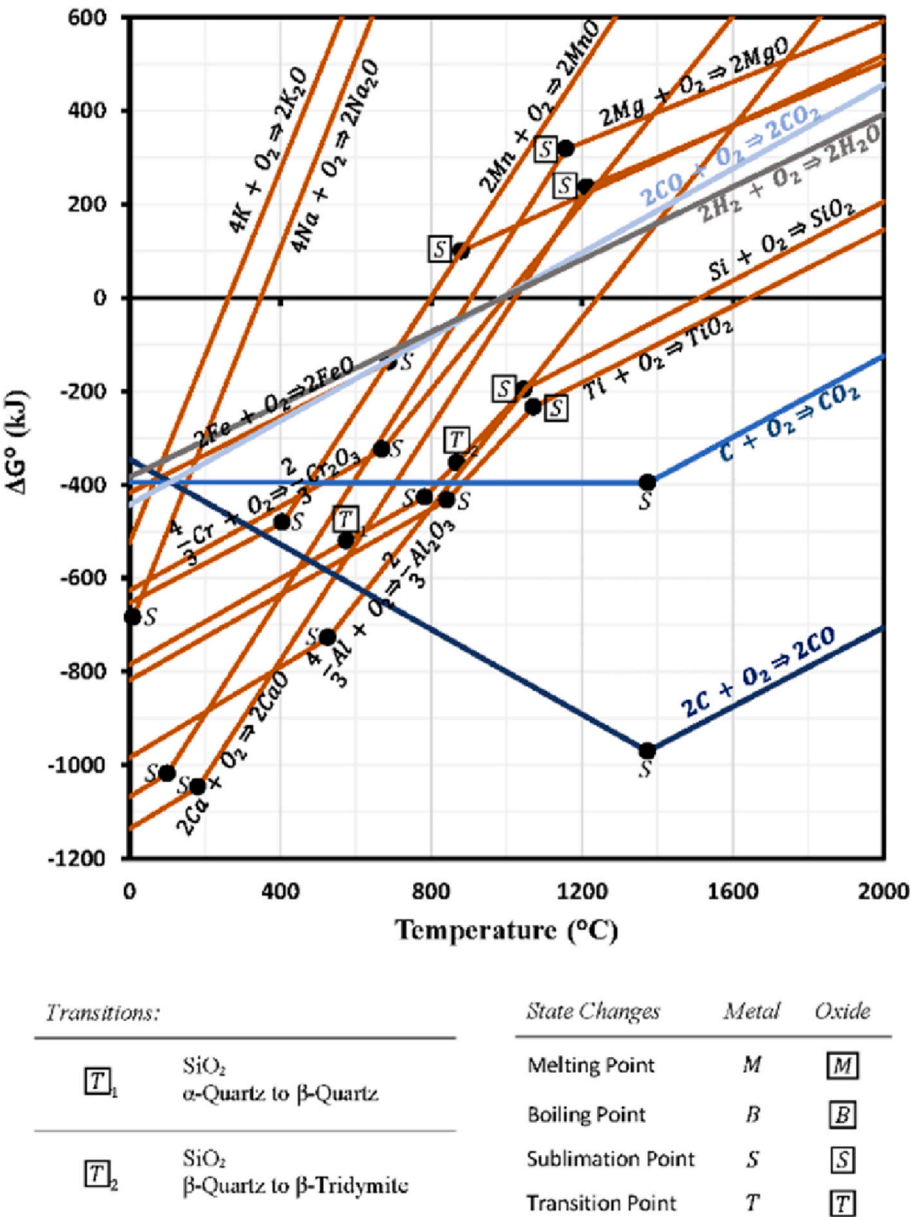


Fig. 2. Ellingham diagram at 3×10^{-15} bar [26].

the generation of oxygen. The overall process occurs as follows:



The reactor design is greatly influenced by the reaction temperature, with two regimes of reduction possible, below and above the melting point of regolith (approximately 1150 °C but influenced by regolith composition). Solid-state reduction of regolith simplifies the reactor design and operation, as the reaction will occur between 1000 °C and 1100 °C at atmospheric pressure, just below the melting point of regolith. This avoids also the complexity of handling molten regolith, and terrestrial reactor designs, such as fixed or fluidized bed reactors, could be adapted. However, due to the lower temperature required for solid-state reduction, conversion will be limited. Troisi et al. [28] performed numerical modeling to simulate the solid-state carbothermal reduction, estimating a theoretical oxygen mass yield of 2.4 %–7.9 %. Higher temperatures allow for the reduction of a higher proportion of the regolith, thus increasing the theoretical yield of the reaction as carbon is able to reduce more species (this trend is observed in Fig. 1). According to Gustafson et al. [29], SiO₂ can be reduced at a temperature of 1650 °C, where the regolith is fully melted, thus permitting higher yields up to 28 %. However, such high temperatures require the complex handling of the corrosive molten regolith. Partial melting has been proposed to use the solid regolith as an insulator.

Solid C was also considered as a reducer [8], and due to the decomposition of CH₄ to C and H₂, the reactivity of solid C is similar to CH₄. An electrolysis cell has been proposed to regenerate the carbon from CO and to produce oxygen. Solid C brings both advantages and disadvantages. A loop process would require extracting solid C from the electrolyzer membrane and mixing it with new raw regolith inside the reactor, which could be challenging to automate. However, shipping a solid from Earth is easier than shipping a gas. In addition, plastic and organic wastes from human activity could also be revalorized as a carbon source. A compromise could be to kickstart the reduction process with solid C while sourcing H₂ locally from lunar water electrolysis to then produce methane with CO and H₂ by Sabatier reaction.

The last possible reducer is CO [8], but it is the least studied one. The process yields metals and CO₂, with CO regenerated from CO₂ by an electrolysis cell or by a catalytic reaction. However, CO can only reduce FeO species according to Fig. 1, limiting the oxygen yield similarly to hydrogen reduction.

3.2.3. Hydrogen reduction

As described earlier, hydrogen reduction involves reacting hydrogen gas with regolith. However, it is important to note that hydrogen can only reduce iron oxides, as shown in Fig. 1. This process is also used for direct reduced iron production as an alternative green process for steel/iron making [30]. When in contact with H₂, the FeO in the regolith is reduced, producing H₂O, which can then be electrolyzed to produce H₂ and O₂. The H₂ is reinjected back into the reactor, closing the loop. The following equations describe the process in the case of FeTiO₃:



The process offers several advantages, particularly in its simplicity. The reduction reaction takes place in one step and requires moderate temperatures of 500 °C–1000 °C at atmospheric pressure in Fig. 1. Although reduction reactions are favored by low pressures, it cannot be envisaged considering the negative impact of low pressure on the kinetics and the high excess of hydrogen required for the reaction. Gas-particle interactions are well known, and reactor designs already exist on Earth, such as particle bed reactors. The products of the reaction are

easy to handle, as the water leaves the reactor as vapor, which can then be condensed and electrolyzed. Ultimately, the process yield will be capped by the ilmenite content of the feedstock, which is approximately 10 % by weight in the best-case scenario with high-Ti Lunar Mare basalts [6]. Adding an upstream beneficiation step can increase the Fe content of the feedstock [31]. The process requires hydrogen to be shipped from Earth to start and to compensate for losses, which can be challenging. However, H₂ could also be obtained by electrolyzing H₂O lunar water present at the poles. Recently, the reaction was performed using endogenous hydrogen directly present inside the lunar regolith [32]. A yield of 5 % was achieved by heating lunar regolith samples from Change'E-5 mission at a temperature above 1200 °C.

3.2.4. Vacuum pyrolysis

Vapor phase pyrolysis consists in exposing the oxides present inside the regolith to the high temperature and low pressure, causing their vaporization in order to achieve a thermal reduction [33]. The reaction produces oxygen and gaseous metals (or suboxides). The vapor phase pyrolysis process only consists of a thermal reduction without any reducer and, therefore, can be represented by Equation (3). At standard temperature and pressure, the thermal reduction cannot occur spontaneously. By increasing the temperature and decreasing the pressure, the Gibb's free energy value can be brought to negative value, permitting the spontaneous reduction of the oxide. Relationship between pressure, temperature, and oxide spontaneously reduced can be seen in Figs. 1 and 2. The equilibrium constant and the Gibbs free energy are linked by the following equation (11).

$$K = \frac{p(M)^x \cdot p(O_2)^{\frac{y}{2}}}{a(MxOy)} = e^{\left(\frac{-\Delta G}{RT}\right)} \quad (11)$$

By following Le Chatelier principle, removing constantly the products avoids the settling of an equilibrium and drives forward the thermal dissociation reaction. Metals and suboxides will be condensed and oxygen will be pumped out of the reactor. Therefore, this process will also permit to limit the reverse recombination reaction (oxidation of the vaporized metals).

According to Figs. 1 and 2, lowering the pressure permits to reduce the temperature required to achieve oxide dissociation. Equilibrium calculations demonstrated that an O₂ mass yield of approximately 10 % can be achieved at 2300 °C and 10^{−3} bar, whereas at lunar pressure a similar yield could be obtained below 1000 °C [34]. High temperatures can be achieved by concentrating solar energy or by laser heating. Direct normal irradiance on the Moon surface is about 1361 W/m², which is 30 % higher than on the Earth due to the absence of atmosphere. Furthermore, a lunar day lasts 29.5 times the earth days, permitting longer operation and limiting heating/cooling cycles. Vacuum pyrolysis of regolith does not require the use of any additional reactants, avoiding shipping from Earth. The process can use any type of raw regolith without beneficiation and could permit high oxygen yield. This under-explored process is discussed in more details in the following section in light of previous experimental studies.

4. Vacuum pyrolysis experiments

4.1. The beginnings and historical background

Regolith was pyrolyzed for the first time in 1971 [35] to study the vaporization process. Knudsen effusion-mass spectrometry was used to characterize two samples from Apollo 12. A Knudsen cell heated by radiation and electron bombardment was used to study the vapor of low-volatility materials, coupled with a mass spectrometer to characterize the species vaporized. Sample analysis was carried out at a temperature up to 2226 °C. The lunar regolith samples 12022 and 12065 were vaporized and partial pressure evolutions were monitored by mass spectrometry. Fig. 3 shows the behavior of the vaporization of the

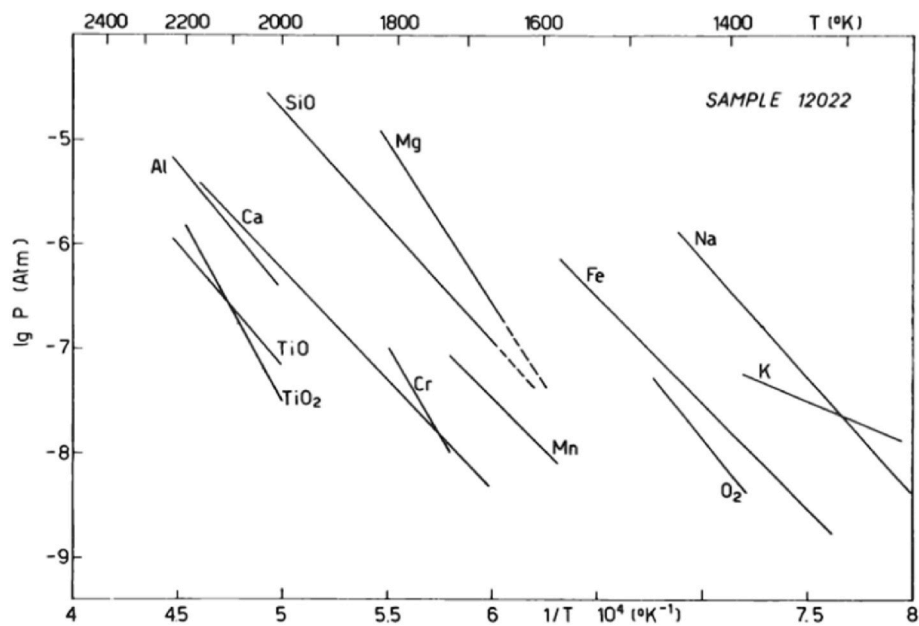


Fig. 3. Vaporization behavior of 12022 lunar sample [35].

sample 12022. As the temperature increased, higher partial pressures of reduced species were observed. Alkaline metals were first observed at low temperatures below 926 °C and 1226 °C, and then Fe and O at temperatures between 1126 °C and 1326 °C. The more refractory species including Mn, Mg, C, SiO, Ca, Al, TiO and TiO₂ were observed at much higher temperatures between 1426 °C and 2126 °C (Fig. 3). Both monoatomic and diatomic oxygen species were detected.

In a report published in 1982, Steurer [33] proposed for the first time the concept of distilling the lunar regolith to produce metals. This process required the regolith to be heated and vaporized by induction. Then, the gas flowed through a series of plates cooled at different

temperatures. The species were separated due to the different condensation temperatures of the metals. Oxygen could be recovered or vented. However, at that time no experiments had yet been carried out.

In 1983, the direction of the project shifted, and instead of metals, the focus was more related to oxygen recovery. Steurer and Nerad [36] calculated the equilibrium at different temperatures of the main oxides present inside the regolith such as TiO₂, SiO₂, FeO, and Al₂O₃. Equilibrium calculations aimed to obtain a partial pressure of the reduced species. A theoretical oxygen yield of about 50 % could be achieved in the range from 1452 °C to 2452 °C for TiO₂, SiO₂, and Al₂O₃. FeO was identified as an exception because it did not yield oxygen compared to

Table 2
Table summarizing the various vacuum pyrolysis experiments by date
* Value extrapolated from the data available in the original communication.

Reference	Year	Institution	Country	Heating mode	Power density (W/cm ²)	Species	Temperature (°C)		Pressure (bar)	
							Min	Max	Min	Max
Steurer	1985	NASA	USA	Induction	Irrelevant	SiO ₂	2375	2375	Unknown	Unknown
Senior	1991	Physical Science (NASA Contractor)	USA	Solar	338	FeTiO ₃	1926	2126	5.30E-04	2.00E-04
						CaAl ₂ SiO ₈	1566	1766	4.00E-04	2.00E-04
Sauerborn	2005	DLR	Germany	Solar	165*	CaO	1640	1640	6.10E-05	1.40E-07
						SiO ₂	1440	1440	3.50E-05	9.70E-08
						ZnO	1190	1190	2.70E-05	1.70E-07
						MS1	1550	1550	8.50E-05	9.10E-06
Matchett	2006	NASA	USA	Solar	365*	FeTiO ₃	1436	1867	9.30E-05	1.30E-03
						MgSiO ₃	1072	1100	1.33E-05	3.10E-04
						Al ₂ O ₃	1800	1800	4.00E-04	4.10E-03
						MLS-1a	1474	1474	1.87E-04	1.87E-04
Tanaka	2019	University of Tokyo	Japan	Laser	51-260	Al ₂ O ₃	2826	3426	1.22E+00	4.05E-01
Shaw	2022	CISRO	Australia	Solar simulator Furnace	132	FeO/Fe ₂ O ₃	1025	1127	3.47E-05	2.77E-07
						FeTiO ₃	1116	1116	2.50E-05	2.50E-08
						LMS-1	1121	1151	8.96E-05	3.88E-08
Li	2022	Kunming University	China	Laser	Not reported	SiO ₂	4700	4700	1.00E-08	1.00E-07
Zhao	2023	Shanghai University	China	Laser	1240-3100	TiO ₂	1948	2278	1.00E-08	1.00E-08
Li	2023	Shanghai University	China	Laser	5969	Fe ₂ O ₃	Unknown	Unknown	1.00E-08	1.00E-08
					3979	FeTiO ₃ /Fe ₂ O ₃ /SiO ₂	Unknown	Unknown	1.00E-08	1.00E-08
					9948	Regolith Simulant	Unknown	Unknown	1.00E-09	1.00E-09
Šeško	2024	ESA	Germany	Solar	Not reported	EAC-1	1476	1626	1.00E-10	1.00E-05
Li	2024	Shanghai University	China	Laser	Not reported	FeO	1726	2176	1.00E-08	1.00E-08

the other oxides. The same study was carried out for a mixture of oxides with a composition similar to the lunar basalt. The oxygen yield by weight for the mixture was about 17 %. A beneficiation step was described involving a magnetic removal system, and as such, FeO could be discarded from the feedstock. It increased oxygen production by 3 %, resulting in a 20 % oxygen yield for an energy consumption of 35 MWh per ton of regolith. Experiments to validate the concept were carried out on single oxides [37]. Only silicon dioxide was tested as it is the main component of regolith. Placed inside the chamber, samples were pyrolyzed by susceptor heating. Volatile species condensed on a cold metallic finger, and silicon and suboxides were recovered. Further investigations were done by analyzing the gas produced. Steurer's experiment confirmed the feasibility of generating oxygen from silicon dioxide by vacuum pyrolysis and paved the way for a potential utilization of lunar regolith as an oxygen source. Subsequently, the following works about pyrolysis mainly focused on using two different heating solutions: either laser or concentrated solar energy. The different pyrolysis experiments with their different parameters such as the heating mode, the species, the temperature and pressure are listed in Table 2.

4.2. Laser vacuum pyrolysis

In this section, the work conducted on the laser pyrolysis is presented. The approach still consists of the dissociation of oxides driven by heat under vacuum. However, the heating is ensured by laser, which has the advantage of delivering very high temperatures and being very convenient. Various species were studied with the laser pyrolysis including Al_2O_3 , TiO_2 , SiO_2 , FeO, and in a lesser extent regolith simulant.

Tanaka et al. [38] developed a method to reduce alumina using laser heating. The laser ablation technique involves vaporizing the surface of a material, with the energy from the laser permitting the reduction reaction. Sample irradiation was achieved using a 400 W–2000 W CO_2 laser. The spot size radius was 0.5 mm, and the power density ranged from 51 kW/cm^2 to 260 kW/cm^2 . Air was first removed from the chamber using a rotary pump and then the reactor was pressurized with argon to 0.405 bar, 0.659 bar, and 1.22 bar. Alumina was sintered into a rod to be used as the laser target. Materials were ejected by the laser, forming an ablation plume. Emission spectrometry was used to analyze the ejecta and to observe the products of the reaction. Al and AlO were detected in the ejection plume, confirming the dissociation of alumina. At a higher pressure and laser intensity, the quantity of reduced species was higher in the ablation plume. An equilibrium temperature was estimated using Planck's law, ranging from 2826 °C to 3426 °C. A higher pressure increased the boiling point of alumina, allowing solid and liquid alumina to absorb laser energy more efficiently than gaseous alumina, thus resulting in a higher degree of dissociation. Further investigations focused on aluminum production through alumina reduction. Tanaka et al. [39] observed spherical aluminum particles on the surface of the rod using SEM (scanning electron microscopy) and EDX (Energy-dispersive X-ray spectroscopy). The effects of preheating the sample by the laser and spot size were studied [40]. Preheating demonstrated higher plume temperatures of up to 4100 K, which resulted in an increased molar reduction. Tanaka et al. [41] found that laser-exposed alumina created oxygen-deficient liquid alumina ($\text{Al}_2\text{O}_x < 2.4$). Upon cooling, this area partially regained oxygen, forming $\text{Al}_2\text{O}_{2.4-2.8}$, with Al particles appearing on the surface. Higher power density increased aluminum particle formation due to higher dissociation. These particles formed at surface temperatures between 933 K and 2300 K, leading to the suggestion of heating the condensation plate to this temperature range to harvest Al formation. Laser ablation was performed on oxygen-deficient samples ($\text{Al}_2\text{O}_x < 2.4$), generating 1.7 times more aluminum and, consequently, more oxygen.

Zhao et al. [42] proposed a method called laser thermal vacuum metallurgy, capable of reducing the TiO_2 present in regolith to produce oxygen. The titanium-oxygen binary system can form lower valency

metastable suboxides when reduced, such as Ti_8O_{15} , Ti_7O_{13} , Ti_6O_{11} , Ti_5O_9 , Ti_4O_7 , and Ti_3O_5 . Thermodynamic analysis was carried out to determine the Gibbs free energy of reduction as a function of the temperature for the above-mentioned species. At 1×10^{-8} bar, TiO_2 started to be reduced spontaneously into lower valence suboxide at around 1700 °C, and TiO formation only occurred above 2152 °C. The sample was heated by a 6000 W ytterbium laser. A spherical vacuum chamber was used and vacuumed by a turbomolecular pump and a scroll pump mounted in series. The sample temperature was measured by a pyrometer. The experimental setup is represented in Fig. 4. Anatase phase (99.0 % titanium oxide) powder was used, and 300 mg of powder was first converted into rutile by being exposed under ambient pressure to the laser for 10 s at 0.62 kW/cm^2 . Under a vacuum of 1×10^{-8} bar, TiO_2 (rutile) was incrementally exposed to a higher power density of 1240 W/cm^2 , 1860 W/cm^2 , 2480 W/cm^2 and 3100 W/cm^2 , respectively. The temperature increased from 1948 °C to 2278 °C with respect to power density. XRD analysis was utilized to determine the phase composition. As the power density increased, the overall valence of the sample decreased, confirming a higher degree of dissociation. Suboxides acted as intermediary steps, creating a pathway to the formation of lower valency titanium suboxide species. The increased temperature allowed the sample to release more oxygen, confirming the thermodynamic analysis predictions. Nevertheless, no TiO was observed in the sample even at the highest temperature.

The process of thermal reduction of SiO_2 by laser was theoretically and experimentally studied [43]. Natural quartz was used as a source of SiO_2 . The sample was cut into 1 cm^2 sheets with an average thickness of 358 μm and dehydrated. Experiments were conducted with a 532 nm Nd-YAG laser at 75 mJ inside a vacuum chamber evacuated to a pressure of 10^{-7} bar to 10^{-8} bar. When the sample was irradiated, an increase in pressure was observed, reaching 10^{-2} bar to 10^{-3} bar. To avoid over-pressurization of the chamber, laser impulsion was used at frequency of 0.2 Hz to allow the chamber to stabilize to 10^{-9} bar. A series of 50 heating cycles was performed. The heating caused the formation of a plume of hot materials, and the vaporized particles were recovered on a copper collecting net. Spot size was measured to be 650 μm by SEM, and the power density was calculated to be 22.6 W/cm^2 . EDX analysis carried out on the collected materials showed a silicon-oxygen molar ratio ranging from 1.74 to 0.12, confirming the feasibility of decomposing SiO_2 by laser. An approximate temperature at the focal point was determined with COMSOL computer software, indicating that SiO_2 could reach 4626 °C.

Similarly to the reduction procedure of TiO_2 , Li et al. [44] patented a

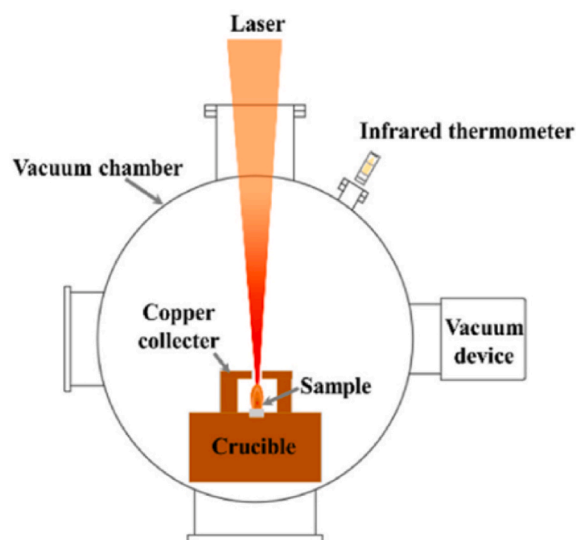


Fig. 4. Schematic of the reactor for laser pyrolysis used by Li et al. [45].

reducing agent-free vacuum iron production method. The technique was showcased inside the patent as follows: iron minerals were exposed to a vacuum of 10^{-4} bar to 10^{-15} bar inside a vacuum chamber and to a light beam such as a laser or concentrated solar energy with a power density in the range of 0.01 kW/cm^2 to 1 kW/cm^2 . A cooled copper collector was placed 1 cm–500 cm from the sample and used to recover some of the iron species. Results obtained using the previously described apparatus were included in the patent. Two samples with mineral species were tested, including pure Fe_2O_3 and a mixture of FeTiO_3 (74.9 %), Fe_2O_3 (14.2 %), SiO_2 (7.9 %), the rest being impurities. Lastly, a regolith simulant was tested with a composition of SiO_2 (45.5 %), Al_2O_3 (17.3 %), FeO (13.2 %), CaO (8.3 %), MgO (5.6 %), Na_2O (2.7 %), TiO (2.6 %), K_2O (1.9 %), P_2O_5 (0.5 %), MnO (0.3 %), with the rest being impurities. A mechanical pump and a turbomolecular pump were used to evacuate the chamber to 10^{-8} bar for the first two samples and for the first two samples and 10^{-9} bar for the last one.

10^{-9} bar for the last one. One gram of the feedstock was placed inside a copper crucible with an 8 mm cavity, and water-cooled at 25°C . A copper disk, 80 mm in diameter and 40 mm thick with a hole of 5 mm in diameter placed at the center, water-cooled at 5°C , was placed 50 mm above the sample and used as a collector. Experiments were conducted using a 6000 W laser with a wavelength of $1080 \pm 5 \text{ nm}$ and a spot diameter of $200 \mu\text{m}$. The laser was defocused to expose the entire surface of the sample. The species were first preheated and then exposed to the nominal experimental power. The gas produced during the process was recovered inside a collection bag for further analysis. Experimental conditions are summarized in Table 3.

The condensed materials on the disc were recovered. The molar percentage content of the powder was determined using EDX and the Fe particles present were also examined with SEM to determine their compositions. The phases present in the first two samples were determined by X-ray diffraction (XRD). The purity of the collected oxygen was determined by a non-disclosed technique. Part of the material was ground and magnetically separated, yielding an iron powder. This powder was analyzed to determine its Fe purity. Table 4 summarizes the results obtained from the experiments.

Elemental Fe formation was revealed by XRD in the deposited materials of the Fe_2O_3 sample. The same analysis for the second sample showed that the involved phases were Fe, FeO, FeTiO_3 , FeSiO_4 and SiO_2 . The results for the regolith simulant were probably inconclusive, as too many phases were present, making it impossible to identify the peaks. The lower purity of the iron powder obtained from the regolith simulant was explained by the low iron content of the raw material and the presence of a greater number of nanoparticles in the collected materials. This patent demonstrated the potential of laser vacuum pyrolysis to produce metals and oxygen from lunar regolith.

After patenting their method, Li et al. [45] published some results on the reduction of FeO to produce metallic iron and oxygen. The experimental setup was very similar to the one used for TiO_2 reduction. Gibbs energy minimization calculation was conducted on FeO utilizing FactSage 8.2 software. The experiment still consisted of a vacuum chamber and 6000 W laser, similar to the one being used previously [44]. Samples were placed in a hemispherical copper crucible (diameter: 12 mm, outer diameter: 80 mm, and height: 30 mm). FeO was first obtained from Fe_2O_3 , then pyrolyzed using the following procedure.

Pellets of 0.5 g were first heated during 6 min at 200 W with laser to form pure Fe_3O_4 . FeO was then formed by a second laser heating during 40 s at a power of 600 W and a pressure of 1×10^{-8} bar. The pyrolysis reaction was directly performed after FeO formation during 20 s at a power of 2000 W. The temperature of the sample evolved between 1726°C and 2176°C during the pyrolysis. Metallic Fe, FeO and Fe_3O_4 were observed by XRD analysis on the copper plate. A content of 12.5 wt % of metallic Fe was determined by titration and a reduction rate of FeO to Fe was determined to be $\sim 15.5\%$. To confirm the origin of Fe_3O_4 , a second identical procedure was proposed where pure FeO was used instead of preparing it. The sample was pyrolyzed for 20 s at 2000 W under a vacuum of 1×10^{-8} bar. It was demonstrated that the composition of the phase is identical between the two-procedures, confirming that the production of Fe_3O_4 originated from the pyrolysis reaction due to the oxidation of FeO. Different morphologies of metallic Fe were observed on the copper plate: spherical nanoparticles, strip like and irregular aggregates. Reduction of FeO to form Fe was demonstrated using laser pyrolysis. Unfortunately, in contrast to the patent, no oxygen production measurements were attempted. In comparison with the patent, the publication provides less optimistic experimental results.

The previously presented studies achieved the reduction of most of the important oxides composing the lunar regolith. Respectively, Al_2O_3 , TiO_2 , SiO_2 , and Fe_2O_3 were dissociated by exposing them to a laser beam, and these oxides account for approximately 85 % of the regolith mass. However, one predominant oxide, CaO, which makes up to 10 % of the lunar soil, has not yet been studied. This oxide remains not considered likely because it is not the most abundant one. CaO is also the most stable oxide in the regolith, requiring more energy and higher temperatures to dissociate it. Nevertheless, the necessary energy levels could be obtained, considering that temperatures as high as 5000°C can be reached by laser heating. However, the laser technology may not be the most efficient option for being used on the Moon's surface, because it requires converting light to electricity via solar cells and then converting the electricity back to light. An optimistic energy yield for the system would be around 12 %, with CO_2 lasers having an efficiency of 30 % and space solar cells yielding 40 %. It is also noteworthy that solar cells tend to lose efficiency over time due to ionizing radiation. Moreover, scaling up the process could present significant challenges, as pilot-scale studies of laser pyrolysis reactors have not been conducted, and the laser spot size tends to remain relatively small. Another arising solution is to directly concentrate solar radiation using an optical system, such as a parabola or a Fresnel lens concentrator.

4.3. Solar vacuum pyrolysis

Solar pyrolysis utilizes the Sun's energy directly, which is abundant and virtually unlimited on the lunar surface during the day periods. Therefore, this section showcases the various studies that have been conducted with concentrated solar energy. Solar vacuum pyrolysis started being investigated long time before laser, although the work is more discontinuous and can be separated by several years. Experiments were conducted mostly on regolith simulants although some oxides or minerals were tested.

Senior [46] was one of the first to address solar pyrolysis of regolith in 1991. She simulated the thermodynamic behavior of the reaction, by

Table 3
Reported experimental conditions (power density was calculated according to available data) [44].

Samples	Distance of the collector to the crucible (mm)	Exposure angle ($^\circ$)	Pressure (bar)	Preheating power (W)	Preheating duration (s)	Heating power (W)	Power density (W/cm^2)	Heating duration (s)
Fe_2O_3	50	Unreported	10^{-8}	300	60	3000	5969	20
FeTiO_3 Fe_2O_3	30	45	10^{-8}	500	20	2000	3979	10
SiO_2								
Regolith Simulant	30	60	10^{-9}	500	20	5000	9948	5

Table 4
Results obtained after the analysis of the pyrolyzed samples [44].

Samples	Fe content in deposited materials (wt%)	Fe particles composition (%mol)	Oxygen purity	Iron powder purity after separation
Fe ₂ O ₃	27.1 %	Fe 98.5 %, O 1.5 %	99.5 %	90 %
FeTiO ₃ , Fe ₂ O ₃ , SiO ₂	30 %	Fe 90.5 %, O 7.2 %, Si 1.3 %, Ti 1.0 %	99.0 %	85 %
Regolith Simulant	Unreported	Fe 92.2 %, O 2.4 %, Si 3.9 %, Ti 1.5 %	98.5 %	45 %

using the Gibbs free energy minimization method. SOLGAMIX program was used to determine the equilibrium composition by minimizing the Gibbs free energy of the whole system. Molten regolith, due to its composition, does not behave like an ideal liquid. To account for non-ideal liquid phenomena, an Ideal Mixing of Complex Components solution model was used. The partial pressure of vaporized species was calculated as a function of the temperature between 1326 and 2126K at a pressure of 2×10^{-4} bar. FeTiO₃ and CaAl₂Si₂O₈ were separately computed to be further compared with experimental data. Equilibrium calculations found out that FeO reduction was responsible for most of the oxygen produced. Despite being the predominant specie in CaAl₂Si₂O₈, silicon dioxide did not contribute significantly to oxygen production. FeTiO₃ released more oxygen than CaAl₂Si₂O₈. Minerals with high FeO content appeared to produce more oxygen and to be more appropriate for oxygen production. The use of high temperatures was more suitable for a high throughput of vaporized materials, nevertheless dissociation of oxygen occurred above 1726 °C. Monoatomic oxygen behavior is still uncertain and has not been accounted as it is more susceptible to react with metals and suboxides than forming diatomic oxygen. Temperatures between 1726 °C and 1926 °C were best suited to produce diatomic oxygen. The reduced species at those temperatures were in majority FeO, SiO, and MgO. The same methodology was applied to analyze thermodynamic equilibrium of a composition similar to the mare soil [47]. Conclusions were close to the modeling conducted with FeTiO₃ and CaAl₂Si₂O₈, since the following species, O₂, SiO, O, Fe, Mg, FeO, and SiO₂, were found in the gas phase. The most abundant product in the gas phase was O₂ produced by the reduction of FeO. Contrary to calculations of [36], where FeO was supposedly not reduced, Senior pointed out that FeO was the main oxygen producer.

Experiments were carried out to compare with the data collected during simulation [46]. Both FeTiO₃ and CaAl₂Si₂O₈ were tested. A vacuum chamber was designed to pyrolyze mineral samples (Fig. 5). A Fresnel lens was used to concentrate sunlight, and samples were exposed to a heat flux of 388 W/cm². Inside the chamber, a condensing cooled copper plate was used to recover the vaporized metals and avoid oxygen recombination. A K-type thermocouple and an absolute pressure transducer monitored the gas temperature and pressure. A sample holder made of platinum foil was used to load 0.5 g of mineral samples.

The chamber was evacuated until the pressure reached between 2×10^{-4} and 5.3×10^{-4} bar, and was then sealed. Oxygen production was monitored by measuring the evolution of the pressure inside the

chamber. A small reproducible leak of 2.66×10^{-6} bar/min was observed. The following procedure was used to prepare CaAl₂Si₂O₈ samples. Crystals were crushed to obtain a powder, and the material was sieved to obtain a particle size between 125 and 65 μm. CaAl₂Si₂O₈ powder was desiccated overnight at 500 °C in an electrical furnace. FeTiO₃ was sieved below 45 μm particle size. The pressure inside the chamber increased for both samples due to the production of oxygen. An increase of 0.2×10^{-4} bar was reported for FeTiO₃ while the calculated temperature reached between 2200 and 2400K. CaAl₂Si₂O₈ sample produced less oxygen since an increase of only 1.3×10^{-4} bar (0.1 torr) was observed (Fig. 6) for a calculated temperature of 1840–2040K. Samples were analyzed by X-ray diffraction before and after pyrolysis. A decrease in oxygen content was detected in the fused samples. The pressure rise observed for CaAl₂Si₂O₈ during experiments was higher than what was expected by the simulation, and the difference was explained by the complexity of the system.

The work of Senior permitted to better understand the behavior of the reaction and the influence of the multicomponent phase, and was at the origin of the distinction between O and O₂ (thus requiring temperature optimization to avoid O formation) [47]. This work further permitted to demonstrate the possibility of extracting oxygen from minerals and not only from pure oxides.

Sauerborn [48] focused on designing a reactor capable of producing oxygen. Experiments were carried out at the high-flux solar furnace of the German Aerospace Center, capable of concentrating the solar energy with a factor of 5000 and reaching a flux density of 450 W/cm². Samples were placed inside a 34 L chamber and exposed to concentrated solar energy from the top. Vacuum was achieved by a series of pumps. Sample temperatures were measured with a pyrometer, and pressures were monitored using gauges. The composition of the reactor atmosphere was monitored by a quadrupole mass spectrometer (QMS) to quantify the oxygen production during the experiments. Experimental parameters were acquired in real time to be analyzed later (Fig. 7). The apparatus reached temperatures up to 1725 °C, and pressures as low as 10^{-8} bar.

Different samples such as CaO, SiO₂, ZnO, and regolith simulant were tested. The melting of regolith was observed between 1200 °C and

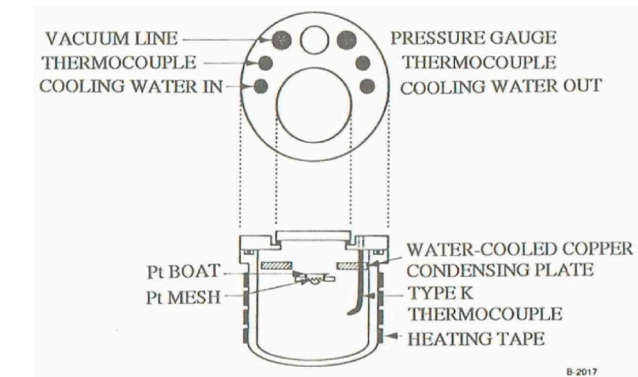


Fig. 5. Reactor used by Senior for the pyrolysis of FeTiO₃ and CaAl₂Si₂O₈ [46].

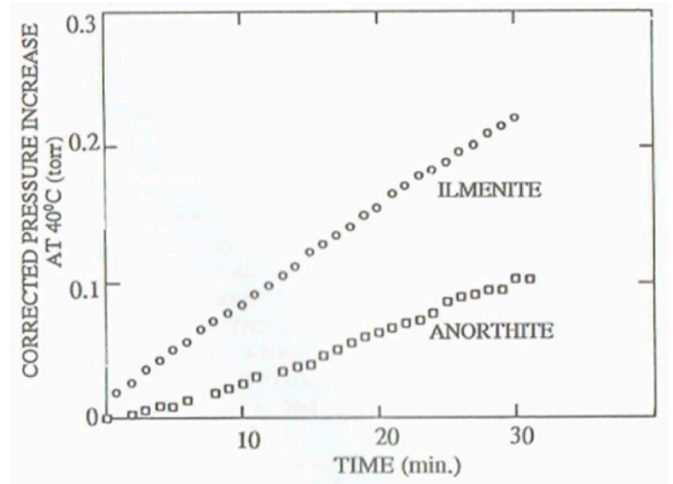


Fig. 6. Pressure increase observed during the pyrolysis of FeTiO₃ (Ilmenite) and CaAl₂Si₂O₈ (Anorthite) [46].

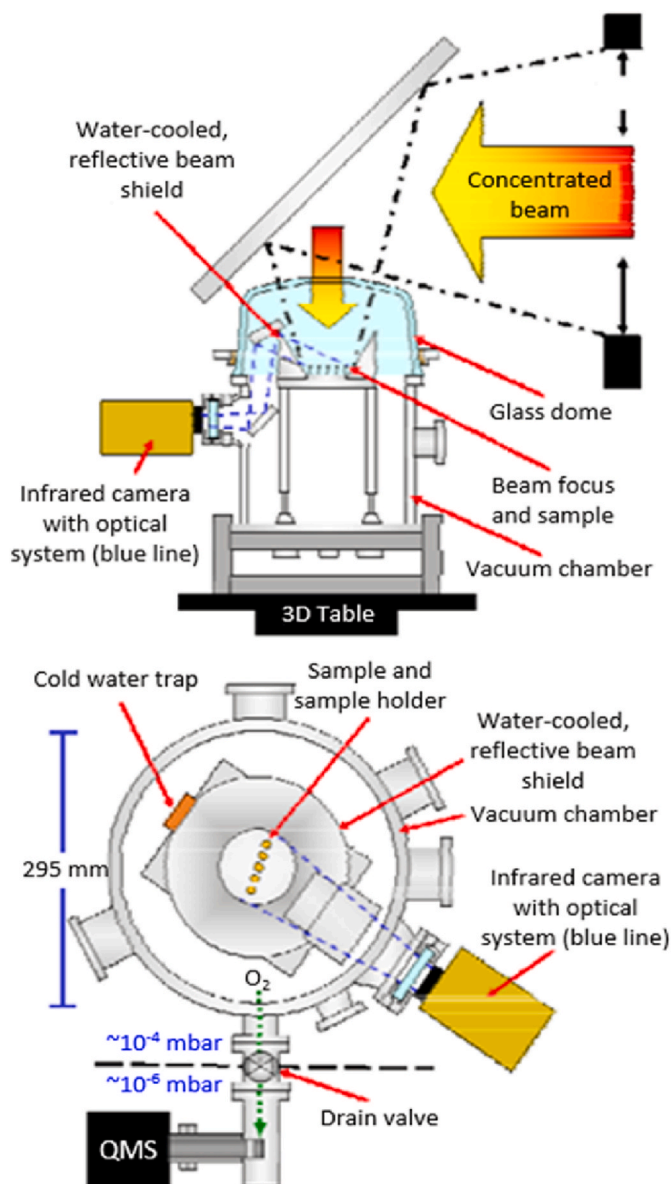


Fig. 7. Schematic of the experimental setup used by Sauerborn (adapted from Ref. [48]).

1300 °C. QMS analysis detected an increase in oxygen partial pressure in the reactor. EDX analysis confirmed a diminution of oxygen content in the samples. This experiment permitted to demonstrate the feasibility of extracting oxygen from a regolith simulant with solar energy concentration. However, the experiment showed some limitations, e.g., the glass dome broke several times due to thermal shock induced by the deposition of metal vapor. Mass spectrometry was also challenging due to overpressure in the chamber.

Equilibrium compositions were computed by using the Gibbs free energy minimization method (HSC Chemistry 5.11 software). Calculations were performed for oxides (SiO_2 , Al_2O_3 , FeO , CaO , MgO , TiO_2), ilmenite, and Apollo 15 regolith composition. As expected, a lower pressure reduced the temperature needed for the dissociation [34]. At 10^{-4} bar, the oxygen production peaked between 1800 and 2100 °C, whereas at 10^{-10} bar, the maximum was reached between 1300 and 1400 °C. A theoretical oxygen mass yield of 14 % could be achieved for a mare regolith composition as stated (Fig. 8).

A small apparatus was subsequently designed by Matchett [34] to perform the pyrolysis reaction on different samples (Fig. 9). Solar concentration was ensured by a Fresnel lens of 1 m² mounted on a one-axis tracking system. The experimental set-up was placed on a wheeled cart.



Fig. 9. Matchett's experimental setup: the reactor can be observed at the center with the Fresnel lens above [34].

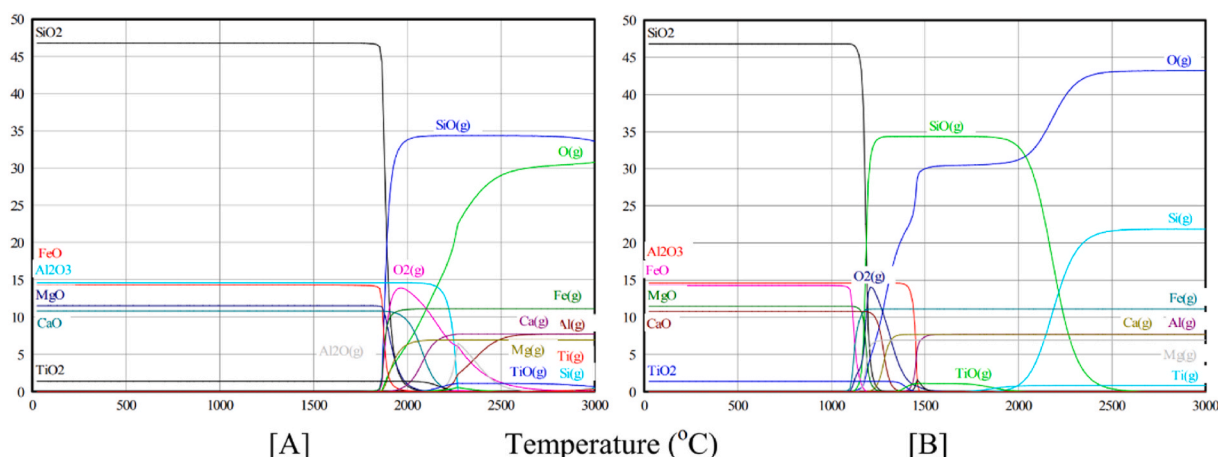


Fig. 8. Thermodynamic equilibrium composition of Apollo 15 regolith pyrolysis at (A) 10^{-4} and (B) 10^{-10} bar [34].

The pressure was lowered with an oil-sealed rotary vane vacuum pump. A scroll pump maintained the vacuum in the chamber while transporting the oxygen produced to the cryocooler. Temperature was measured using C-type thermocouples placed inside the crucible and a handheld pyrometer in case of failure of the thermocouples. The pressure measurement was carried out using a Convectron vacuum transducer. The pyrolysis gas was analyzed in real-time by a residual gas analyzer (RGA). This device required a pressure below the operating pressure inside the reactor. As a result, the gas was collected from the main circuit by a leak valve and, inside a secondary circuit, the pressure was lowered by a turbomolecular pump ensuring proper operating vacuum condition for the analyzer.

Different species were tested with this apparatus and 11 experimental tests were conducted. The samples tested were FeTiO_3 , MgSiO_3 , Al_2O_3 , and MLS-1a regolith simulant. During experiments, the temperature varied between 900 and 1400 °C and peaked at 1900 °C. A mass decrease was observed for most of the samples. The regolith simulant achieved the highest mass loss of approximately 10 %. RGA measurements were only conducted with FeTiO_3 samples and oxygen release was observed but without any indication about its origin. Spectral analysis by SEM/EDX revealed a reduction of oxygen content and an increase in the metallic fraction in the residue. This confirmed the production of oxygen. The experiment duration ranged from less than 1 min to more than 1 h. This work was similar to the one of [48] as they were realized at the same period, and both had to use a secondary circuit to ensure low pressure for their mass spectrometer. Similar issue was encountered with the glass window that broke due to thermal shocks.

Cardiff, who worked with Matchett at NASA, presented a twelfth experimental test during which he was able to measure the oxygen partial pressure with the RGA for a sample of MLS-1A [49]. In this final test, there was an increase in oxygen partial pressure when the simulant began to boil. The oxygen increase was concomitant with a rise in species with an atomic mass of 28, which was likely silicon from MLS-1A. The partial pressure of water remained stable throughout the experiment, indicating that the oxygen was coming from pyrolysis reactions rather than atmospheric sources. No increase in silicon monoxide was reported, suggesting that it either condensed before reaching the RGA or that the oxygen resulted from the reduction of a different species. Furthermore, a second setup was mentioned, designed to vaporize up to 1 kg of regolith using a 3.8 m dish with a flux of 9 kW. However, due to the cancellation of the Constellation program, the reactor was never tested unfortunately, to our knowledge.

Shaw et al. [50] focused on metal recovery, by simulating a multi-stage separation system similar to the principle of distillation.

Thermodynamic equilibria were determined based on Gibbs energy minimization using FactSage 7.11 with FACTPS, FTOxid, and FSstel databases. Two feedstocks were simulated from different lunar regions. The first one concerned maria region from the sample 24999 with high Fe and Mg content and low Na, K and Ca. The second simulation was carried out focusing on highland composition from the sample 64501 with higher concentrations of Ca and Al and low amounts of Fe and Mg. To reproduce operation conditions on the Moon, the pressure was set at 10^{-15} atm. To simulate the separation of the vaporized species, the simulation was broken out into three stages (Fig. 10). Stage 1 replicated the vaporization of the regolith at temperatures between 700 and 1200 °C. Stage 2 was the first separation where the low volatile metals are recovered at a temperature between 0 and 800 °C, and this stage used the composition of the vapor phase of stage 1 at 800 °C. Last stage (stage 3) simulated the separation of the high volatile metals, and it used the composition of the gas phase of stage 2 at 550 °C. Equilibrium was computed at temperatures between –200 and 600 °C. When the temperature of simulation was increased, two phenomena were observed. A greater variety of metals was found inside the gas phase; while monatomic oxygen replaced diatomic one. At 800 °C, stage 1 output was mostly composed of diatomic oxygen, sodium, and potassium. The purpose of stage 2 was to condense the low-volatility species such as Fe. At 550 °C, no Fe was observed, and only alkaline metals were present inside the gas phase. The remaining species were condensed in stage 3. This kind of simulation showed how fractional vaporization of regolith could theoretically separate the different metals inside lunar soil and produce oxygen. However, this kind of setup required a precise temperature control system.

Instead of using vacuum pyrolysis to generate oxygen gas, Shaw [51] used it to produce metals. Experiments were conducted at the Swinburne Solar Simulator (Fig. 11). This equipment can replace a more conventional concentrator such as a parabola or Fresnel lens, and 7 lamps of 6 kW enlightened the reactor. The reactor was a tube of fused quartz equipped with a window to expose the sample to concentrated light. In front of the simulator, an electrical furnace was placed, and the furnace was used to hold the quartz tube. It is important to note that the electrical furnace was only used to preheat the reactor and avoid thermal shock. The tube was depressurized by a turbomolecular pump. Pressure was monitored by a PEN101 cold cathode sensor, and one thermocouple was inserted inside the tube to measure the temperature.

The tested samples ranged from oxides to lunar simulant, passing by mineral samples. The temperature varied from 1025 to 1150 °C and the average pressure from 2.5×10^{-8} bar to 3.47×10^{-7} bar. First, a mix of FeO and Fe_2O_3 was tested, and a colored deposition was observed on the

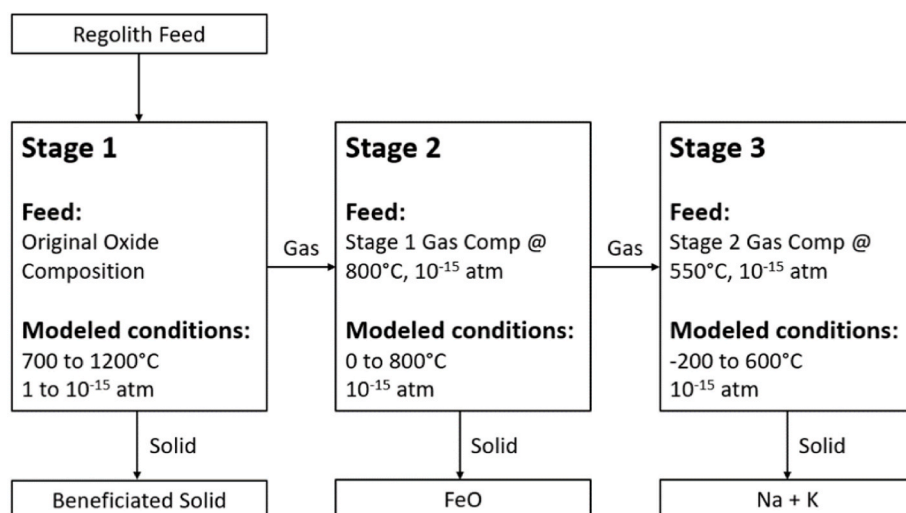


Fig. 10. Diagram of the separation stage adapted from Ref. [50].



Fig. 11. Swinburne Solar Simulator: the quartz tube is placed inside a furnace on the right [51].

surface of the quartz tube. To verify the presence of iron and oxygen, EDX analysis was performed, confirming the sublimation and dissociation of the oxide. A temperature over 1127 °C was needed. XRD analysis was also used to determine the phase composition of the products.

Different colored bands on the tube walls were observed, and these bands correspond to the deposition of the species according to their volatility. The closer the species was to the heat source, the lower it was volatile. EDX analysis showed the presence of impurities such as zinc, lead, chlorine, and potassium. These impurities can be explained by the

fact that the mineral was naturally sourced rather than being a pure chemical species. No titanium and iron were observed on the spectra despite being the majoritarian elements. Regolith simulant samples were tested and multiple deposition bands were observed. The analysis showed many different species, and crystal formations containing Na, Cl, K, S, and O were observed on the tube. Similar to the test with FeTiO_3 , the predominant species were not found on the quartz tube.

Recently, Šeško et al. [52] proposed a similar approach to that of other authors. Their methodology involved a combination of thermodynamic modeling and experiments in a solar vacuum furnace. FactSage 8.1 software was used to model both equilibrium through Gibbs energy minimization and the conditions permitting the elucidation of phase transition temperatures according to pressure. Calculations were conducted to optimize oxygen production across a wide range of temperatures (25–3500 °C) and pressures (10^{-8} to 10^{-3} bar). Several regolith compositions were simulated, including Luna sample 24999, Apollo samples 64501 and 70051, representing Mare, Highland, and High-Ti Mare types, respectively, along with the EAC-1 regolith simulant for comparison with experimental data. The results revealed key phase transitions, with regolith melting between 1076 °C and 1256 °C, and vaporization beginning around 1526 °C at 10^{-6} bar. At a lower pressure of 10^{-8} bar, regolith sublimated directly from solid to gas without passing through a liquid phase. Oxide dissociation followed a specific order, with alkaline oxides such as Na_2O and K_2O dissociating first, followed by FeO , SiO_2 , MgO , and CrO_2 at higher temperatures, and finally TiO_2 and Al_2O_3 . The dissociation order aligned well with previous literature data. Across the entire pressure range, diatomic oxygen ranged between 6.6 % and 14.1 % before full vaporization temperature, with almost no monoatomic oxygen. Regolith composition played a significant role in the efficiency of the process. A comparison of different regolith compositions showed that low-Ti Mare regolith yielded more

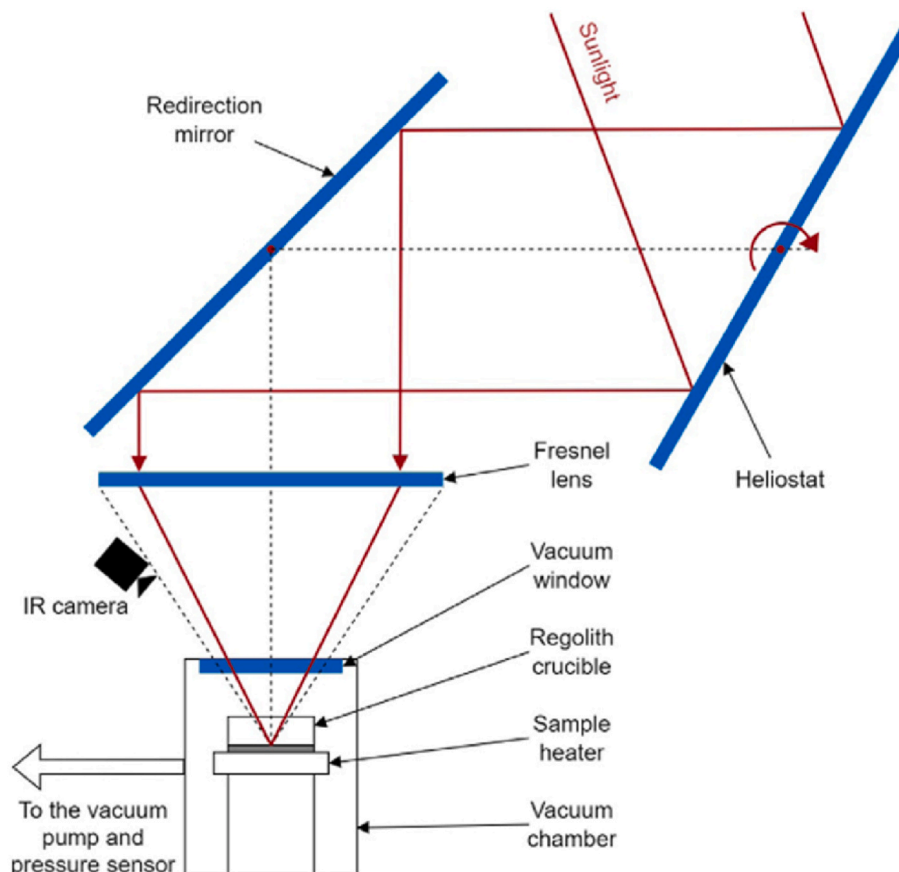


Fig. 12. Schematic of the experimental rig used by Šeško [52].

diatomic oxygen than high-Ti Mare and Highlands regolith. However, when both molecular and atomic oxygen were considered, the Highland composition produced the highest total oxygen yield.

Their vacuum solar furnace setup consisted of a vacuum chamber (Fig. 12) similar to the devices previously developed by Senior [46] and Matchett [34], along with an optical system to concentrate sunlight. Solar energy was focused into the vacuum chamber using a heliostat to track the sun, along with a deflecting mirror to reflect the light onto a horizontal Fresnel lens to heat the regolith. This optical system permitted solar tracking along two axes, horizontal and vertical, while offering the advantage of not having to move the reactor alongside the Fresnel lens. Instead of classical glass, the window was made of quartz glass, likely to avoid the cracking observed previously. Temperature was measured by an Optris PI 1M infrared camera, while pressure was monitored by various pressure sensors.

Experiments were conducted over two days. On the first day, 10 g of the EAC-1A sample was placed in an alumina crucible inside the vacuum chamber, and the pressure was reduced to 10^{-8} bar. The system was then heated to 180 °C, and the crucible to 400 °C, to remove volatiles. Pumping was maintained until the second day and continued during the experiment. The sample was then exposed to concentrated sunlight and heated to temperatures between 1476 °C and 1626 °C, with pressure varying from 10^{-10} bar to 10^{-5} bar throughout the experiment. The heliostat position was adjusted at 5-min intervals to ensure proper concentration on the sample. The experiment resulted in partial melting of the sample, with vapor deposition on the surface of the crucible. The regolith melt exhibited a homogeneous, glassy structure with gas bubbles forming at the base, indicating the release of volatiles during heating. Post-experiment analysis by SEM-EDX imaging revealed the deposition of Fe, Na, and K, consistent with thermodynamic modeling.

Although vacuum pyrolysis has been studied for nearly 40 years, no one has yet been able to experimentally quantify oxygen production, although oxygen was detected by mass spectrometry. A probable explanation could be the inability of setups to achieve sufficiently high temperatures due to rudimentary solar concentration systems. In situ mass spectrometry has proven challenging because of the extremely low pressures required for analysis. Additionally, repeated failures of glass windows due to thermal stress have been observed, further complicating the operation of the reactor. Based on the previous experiments and feedback, it is recommended to develop a solar reactor capable to withstand high temperatures without window breakage, and to quantify the possible oxygen yield from solar regolith pyrolysis. This key aspect is of outmost importance as an experimental assessment of the oxygen mass that can be extracted from a given mass of regolith will enable sizing a solar process to be implemented on the Moon. Continuous operation will be a further step. It is theoretically possible to leverage the vacuum conditions available on the Moon but the oxygen recovery at such a low pressure has to be considered. Selective condensation of metal species is also of interest for lunar soil exploitation beyond the oxygen production from regolith.

4.4. Paving the way toward an integrated lunar process

There is still a long way to go before the oxygen production process can be integrated on the Moon. Different reactors were proposed in the literature and were operated at lab scale. However, most of the proposed designs consisted of a batch reactor, where the sample is loaded inside a small vacuum chamber, and then the sample is heated by concentrated light through a glass window or by a laser beam (or other heating means). While being convenient to demonstrate the feasibility of the process, these reactors cannot ensure high oxygen production because they are limited by the amount of loaded material. A continuous process should thus be considered. Reactors presented here are solar-heated, as the literature lacks of laser pyrolysis reactor examples at a pilot scale. Despite being solar-heated, some concepts could potentially be transferred to laser heating applications. To date, two theoretical designs

have been proposed for continuous operation. The first concept was by Senior [47] with a slide that was used both as a feeding system and reactor bed (Fig. 13).

Steurer [53] proposed the use of a wheel containing the regolith (Fig. 14). The regolith is loaded and then the wheel rotates inside the furnace where the feedstock is exposed to concentrated light. The wheel acts as the crucible and the feeding mechanism.

Recently, this design was improved by Lamboley [54] (Fig. 15). The advantage of the rotating system lies in the fact that it does not require a continuous feeding and discarding mechanism for the pyrolysis step. When the regolith is depleted in oxygen, it can be evacuated, and the system does not have to cope with the physical properties change in the melt and the slag disposal. The feeding system is also slightly simpler, but it requires the crucible to be airtight with the furnace.

Another source of inspiration for solar reactor design could lie in the technologies developed for solar fuels production from two-step thermochemical redox cycles [55–57]. These processes also involving metal oxides aim to split water or carbon dioxide to produce oxygen and hydrogen or carbon monoxide through two sequential chemical reactions. The first step is quite similar to regolith vacuum pyrolysis, as it requires the reduction of metal oxides at high temperatures to release oxygen, which can be achieved using solar energy concentration [58]. The key difference is the product of interest: oxygen in the case of regolith pyrolysis and reduced metallic species in thermochemical cycles. The second step in thermochemical cycles involves the oxidation of the metal (or the reduced oxide) with H_2O or CO_2 to produce H_2 or CO . Suitable oxides are classified into two categories: volatile and non-volatile. Volatile oxides, such as ZnO and SnO_2 , form gaseous reduced species during reduction (Zn or SnO), whereas non-volatile oxides, like CeO_2 and Fe_3O_4 , remain in the condensed phase and mainly involve solid-state reactions [59]. According to thermodynamic simulations, lunar regolith appears to behave closer to a volatile oxide, since all regolith oxides can be vaporized if the reactor operates under sufficiently harsh conditions (low pressure and high temperature). However, regolith may fall at the interface between volatile and non-volatile oxides, because an optimized process might not require full vaporization. Moreover, depending on the pressure, regolith can form a melt, which is not the case for most of oxides used in thermochemical cycles. This is a crucial point for reactor design that requires further study. Recombination with O_2 poses a similar challenge, reducing yields of both oxygen and metals [60]. This issue has been addressed by operating under high vacuum in an inert atmosphere (for ensuring low oxygen partial pressure) or by rapidly quenching the vaporized reduced species. Early experiments on metal oxide reduction were conducted with small quantities in batch mode to understand the mechanisms and parameters of the reaction [61,62]. These experiments involved a batch reactor made of an airtight glass sphere, allowing for controlled

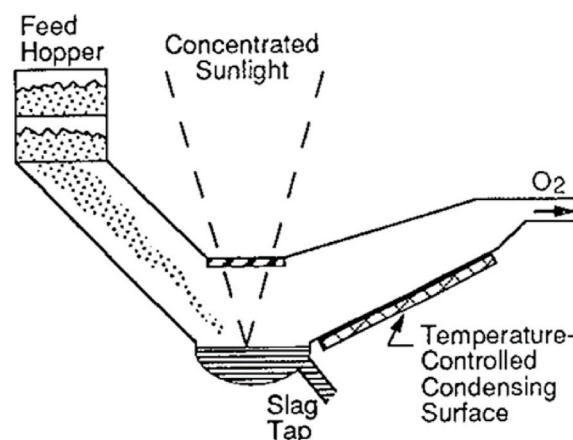


Fig. 13. Continuous solar reactor design [47].

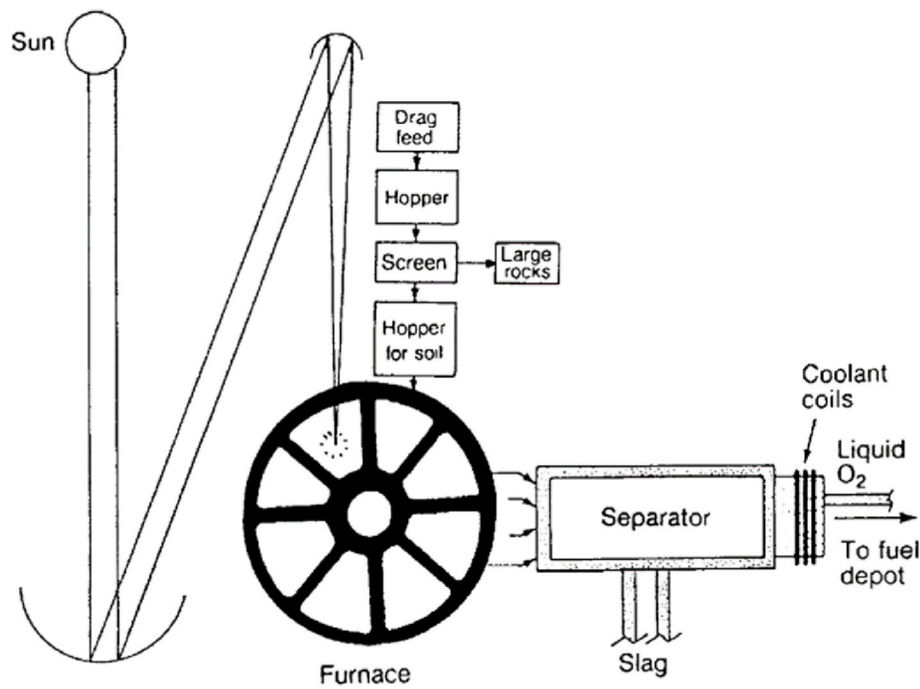


Fig. 14. Wheel mechanism reactor [53].

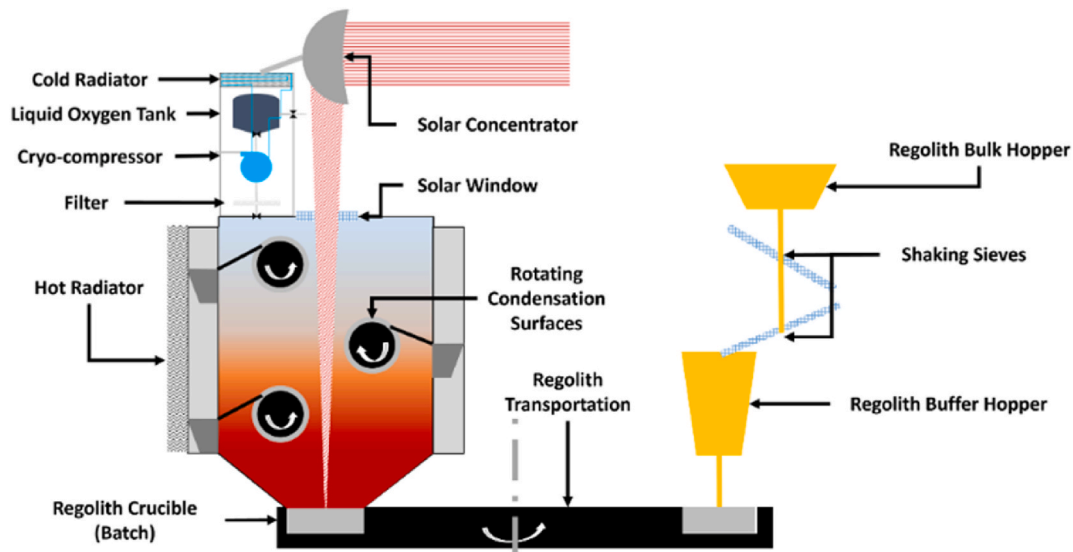


Fig. 15. Conceptual design of rotating furnace by Lamboley [54].

atmosphere and exposure to concentrated solar energy to reach temperatures above 2000 °C [63–65]. Vaporized metal species (e.g., Zn or SnO) were collected on a cold surface or on a filter [60]. In order to increase the production capacity, different continuous particle reactors were designed and experimented. Reactors can be classified according to their suitability for use with volatile or non-volatile oxides [58,66].

Different continuous solar reactor designs for volatile oxides were developed, such as the rotating kiln reactor, which was initially operated for ZnO reduction [67]. These reactors consist of a horizontal rotating cavity equipped with a front window that allows exposure to concentrated solar energy [68]. Oxide particles are injected at the back of the reactor along its axis through the feeding mechanism, and the centripetal force presses the particles against the reactor walls, creating a homogeneous layer. This particle layer serves both as a reactant and as an insulation layer to protect the reactor walls. Recombination is prevented

by injecting inert gas into the cavity, while vaporized metals and oxygen are evacuated along with the quenching gas. Several concepts were developed with varying capacities [67,69,70]. To avoid rotating parts, which complicate operation, a vertical solar irradiated cavity reactor was designed. This reactor consists of an inverted conical structure, with an assembly of 15 hoppers placed around the base of the cone [71]. ZnO powder is dispensed by the hoppers along the walls of the reactor, forming a moving bed of reactant. The reactor outlet consists of two coaxial tubes located at the tip of the cone. The outer tube collects unreacted product, channeled by gravity, as only a fraction of the oxide is reduced. The reaction products are carried by a flow of inert gas and evacuated through the inner tube. A moving front solar reactor was developed to reduce ZnO and SnO₂ pelletized powders, consisting of a refractory cavity equipped with a feeding mechanism to inject oxide pellets into the reactor [72]. Pellets were piled and pushed by a rotary

screw piston. This reactor required the use of an inert gas, with the inlet and outlet placed inside the cavity to lower the recombination rate. Reduced species were recovered on a filter. The reactor allowed for continuous operation at reduced pressure, thanks to the absence of any rotating parts that could cause sealing issues.

Other technologies have been developed for non-volatile oxides; however, they may not be the most suitable due to the vaporization of regolith. Additionally, reactor designs tend to be simpler since the oxides remain in the condensed phase and do not vaporize. The simplest reactor design consists of a packed bed reactor, where the oxide is directly placed inside the reactor [73]. The fluidized bed reactor is an evolution of this concept, where particles are suspended by an inert gas. The particles are transported to the upper region of the reactor, where they are exposed to concentrated solar flux and reduced, before returning to the bottom part of the reactor, creating a convection movement, as in the internally circulating fluidized bed concept [74]. Fluidized beds offer higher heat and mass transfer and better temperature homogeneity compared to packed bed reactors. Other concepts involved monolithic reactors, which use a solid porous structure that acts as both a volumetric radiative absorber, capturing concentrated solar energy, and as the reactor itself. These structures can be made of porous ceramics coated with oxides or directly from the reactive material, such as ceria [75–78].

In the field of oxygen extraction from lunar regolith, several aspects and considerations on the development of the process can be suggested for future applications on the Moon. The technology readiness level (TRL) of vacuum pyrolysis remains low and can currently be considered at TRL 2, where the technology concept and application have been formulated. Achieving TRL 3, which entails an experimental proof-of-concept, is essential for advancing vacuum pyrolysis. Therefore, it is recommended to first confirm the feasibility of extracting oxygen and quantify the potential oxygen yield from the pyrolysis of solar regolith. This step is of utmost importance, as experimentally estimating the mass of oxygen that can be extracted from a given mass of regolith will facilitate accurate sizing of the systems to be implemented on the Moon. In the case of laser heating, the process appears to be nearing demonstration. However, for solar pyrolysis, a reactor must be developed to address the various challenges identified in the literature. Once the feasibility is demonstrated, efforts should focus on maximizing oxygen yield through various approaches. Specifically, it will be crucial to test different types of regolith (e.g., mare, highland) to identify the most suitable resources for the process. In that sense, using existing solar reactors that proved effective operation under high solar flux is recommended. In the case of regolith pyrolysis, the condensed species should finally be analyzed in order to check if it is also possible to recover reduced species of interest for ISRU (metals for example). Moreover, as recombination of oxygen with reduced species may be a strong bottleneck, it is important to study the impact of competing parameters including pressure and temperature.

The literature lacks data on the fundamental chemistry of regolith pyrolysis at high temperatures and its vaporization mechanisms. To complement the above-mentioned points, two techniques (including differential scanning calorimetry (DSC) and the Knudsen cell) could be considered to investigate the thermodynamic and kinetic properties of regolith. DSC enables the study of the thermal behavior of regolith under controlled conditions and provides information on phase transitions, such as melting, crystallization, or decomposition. Applied to the context of pyrolysis, DSC could be used to determine phase change enthalpies and the heat required for reactions. The technique has already been used with various simulants; however, a limitation of these analyses was that they were conducted at ambient pressure instead of under vacuum conditions [79]. A melting point between 1000 °C and 1200 °C was observed depending on the simulant, because compositional aspects are heavily related to phase change temperature. It is likely that the results will be similar from ambient to lunar pressure, due to minimal change in volume between condensed phases, but it needs to be

experimentally confirmed. In parallel, studies should be conducted using a Knudsen cell coupled with a mass spectrometer. This approach allows for the determination of key thermodynamic properties of sublimation, derived from temperature-dependent vapor pressure measurements. Furthermore, the rate of material flux through the cell provides insights into reaction kinetics, enabling the calculation of activation energies. The combination of these two techniques would facilitate a deeper understanding of the physicochemical aspects of regolith reactions across a range of temperatures and pressures.

Advancing the TRL should further involve developing a scaled-up version of the system capable of processing larger quantities of regolith and operating in continuous mode. The initial step should include studying the process on Earth using systems designed to handle greater amounts of regolith and produce higher yields of oxygen. On Earth, continuous operation tends to be the principal choice when designing an industrial process. Adapting the process for implementation to the Moon creates new challenges, and most unit operations will have to be redesigned. A comparative study should be conducted to evaluate the pros and cons of solar and laser heating to determine the optimal heating mode. Laser heating may achieve higher temperatures than solar heating; however, these extreme temperatures may not be necessary for the reaction under vacuum conditions on the Moon, based on the available thermodynamic data. If laser heating is powered by solar cells, an additional energy conversion step is required. Conversely, if it is supplied by nuclear-generated electricity, the process can operate during the extended lunar night. Regarding scalability, larger spot sizes (up to a meter in diameter) are more easily achievable with concentrated solar energy, whereas laser spot diameters are typically limited to a few centimeters. This distinction is crucial, as the choice of the heating mode will significantly influence the reactor's design.

Despite the lack of results from the proposed experiments and feedback, several guidelines can be established for designing a lunar pyrolysis process. Molten regolith is known to be highly corrosive. While using a crucible is feasible, it is essential to identify suitable materials capable of withstanding the extreme temperatures without degradation, such as zirconia or tungsten. The pyrolysis reaction may release highly reactive monoatomic oxygen, a byproduct that should be minimized or avoided. The behavior of monoatomic oxygen remains uncertain; however, it is likely to react with itself (recombining into molecular oxygen), interact with vaporized metals, or corrode the reactor walls. Addressing these challenges will be critical for the successful implementation of the process. Vapors condensation on not intended surface should be avoided especially on optical window and to a lesser extent on reactor walls. In Sauerborn [48] and Matchett [34] experiments, the windows shattered multiple times. The vaporized species condensed on the cold glass, creating a layer absorbing the light with hot spots, and causing overheating, which can damage the window due to thermal shock. To prevent this issue, different techniques are possible. A shielding gas can be used to deflect the pyrolysis gas, similarly to solutions implemented for the thermochemical cycle applications. Alternatively, the glass window could be heated to avoid the condensation of the species, but an appropriate material would have to be used, for example quartz or sapphire glass. A cold surface could also be used to attract the volatile species and avoid deposition on the window. Keeping the glass away from the vapor flux would reduce the probability of deposition. The window could be relocated for example on the side of the reactor and far from the crucible. However, it requires to design a solar concentrator with a high focal length and a small enough spot diameter, so that the radiation passing through the glass window can reach the regolith. Nakamura et al. [80] built an optical waveguide. A cluster of parabolas that concentrated light was used. Then, the solar energy was channeled to the reactor through optical fibers. Laser heating could solve some of these issues. The gas produced was quenched to avoid recombination (usually, a plate condenser is proposed in designs). The condensation process still needs to be studied, in particular the temperature at which the different species condense. Lamboley [54],

inspired by zinc powder manufacturing, proposed the use of rotating cylinders. The powder formed on the surface was then scraped by a blade, while this solution permitted continuous operation. Differential heating of the cylinders would permit the separation of the species according to their condensation temperature. Another method proposed to separate regolith component is by taking advantage of the Marangoni effect. Experiments conducted on JSC-1A demonstrated the upward displacement of the simulant along the crucible wall and change in composition was observed along the vertical axis by SEM-EDX [81,82].

The strength of vacuum pyrolysis lies in its ability to leverage the low lunar pressure, enabling operation at reduced temperatures and, consequently, lowering thermal energy requirements. However, recovering oxygen at such low pressures presents a significant challenge. One approach could involve the use of a cold trap with alternating cooling and heating phases to condense and subsequently release oxygen. Alternatively, oxygen-sorbent materials, such as zeolites, could be explored. Another option is the use of a shielding gas to transport the oxygen produced. However, introducing gas into the reactor would increase pressure, making it difficult to maintain the low-pressure environment needed for efficient oxide dissociation. This approach also raises the temperature requirements and has additional drawbacks: separating the gas from vaporized minerals and released oxygen is complex, and the carrier gas would need to be replenished from Earth due to process losses. A cycling strategy involving intermittent gas injection and heating might mitigate pressure elevation. Once oxygen is recovered, its purification and conditioning should align with other ISRU processes and could be adapted from more conventional technologies. Less critical but still important aspects include regolith supply and upstream processing. Depending on the reactor design and injection system, it may be necessary to remove the coarse fraction of regolith and process only the finer dust. A sifting mechanism could be employed to achieve the desired particle size, with the coarse fraction either discarded or crushed to a suitable size for reuse. Additional considerations include reaction monitoring (e.g., temperature, pressure, oxygen throughput, regolith consumption, and inline analysis), avionics for system control, potential use of concentration systems or heliostats for solar heating, oxygen recovery, purification and conditioning, and overall system integration for lunar deployment.

As conclusive remarks, solar and laser vacuum pyrolysis represents a subset of the technologies in the domain of ISRU, which aims to minimize the need for transporting resources from Earth to the Moon. By utilizing solar energy or laser to extract and process resources on the lunar surface, it reduces reliance on Earth resupply missions and significantly lowers mission costs. This method avoids the need for bulky, complex, and energy-intensive equipment for resource extraction and processing, making it an energy efficient approach. Vacuum pyrolysis can contribute to the sustainability of long-duration lunar missions, and it supports the establishment of lunar bases, enabling extended stays and facilitating future human exploration and scientific research.

5. Conclusion

The lunar regolith contains abundant resources such as metal oxides and volatile compounds. Pyrolysis can be employed to extract and utilize these resources. By subjecting regolith to high temperatures (e.g., using concentrated solar energy), volatile species and oxygen can be released. These resources can then be used for life support systems, propellant production, or manufacturing processes. To date, a limited number of works reports on lunar regolith pyrolysis. In most of the scientific research published on vacuum pyrolysis of regolith, a similar approach was proposed: a thermodynamic simulation was coupled with experimental studies, and the obtained data were compared. Usually, thermodynamic calculations aimed to compute an equilibrium and to determine a theoretical O₂ yield. A strong limitation of this methodology is to consider that the yield corresponds to the point where maximum O₂

is observed at equilibrium. This assumption necessitates considering the reactor as a closed system with homogeneous pressure and temperature conditions, where the composition remains constant and does not evolve throughout the reaction. However, a process might necessitate continuous pumping for the evacuation of the vaporized species, thus shifting the equilibrium. The developed experimental set-ups were also similar, usually consisting of a batch reactor that was vacuumed and then heated with solar energy or by laser. To quantify the oxygen production, two methods were used, either measuring the increase of partial pressure inside the sealed reactor or using a mass spectrometer for produced gas analysis. The tested samples tended to be similar. First, a simple system involving metal oxides was studied, and then it was complexified by testing mineral species to consider the behavior of multicomponent phases. Finally, to approach real operation conditions, regolith simulant was pyrolyzed. When the heating was ensured by concentrated solar energy, the operating temperature did not exceed 2000 °C. Laser heating reached higher temperatures and higher dissociation, probably leading to the formation of monoatomic O (not mentioned in the publications). However, laser heating is less suitable, because electricity must be produced and then converted in a laser beam, whereas abundant and virtually unlimited solar energy can be directly concentrated, and the heat can be used without any intermediate conversion. Vacuum operating pressure varied from 10^{−3} bar to 10^{−9} bar. More recent studies used lower vacuum levels, due to technological evolutions or an easier access to stronger pumps. Experimentally, a stronger vacuum was correlated with a lower pyrolysis temperature at least for solar heating.

All the reported results highlight a loss of sample mass and an increase in oxygen partial pressure, thus confirming the possibility of producing oxygen by vacuum pyrolysis, in agreement with the thermodynamic studies. The interactions between single oxides and multiple phase components remain to be investigated. The influence of regolith composition must also be considered, as it is possible that a feedstock containing more low temperature dissociating oxides favors the formation of diatomic oxygen. Mineral phase behavior has to be investigated as well, and both fractions of unreacted and vaporized materials should be analyzed to identify the species formed. It is also crucial to determine the optimal temperature for the reaction ensuring the highest oxygen throughput while not producing monoatomic oxygen. Continuous operation has also to be explored. Early conceptual designs of continuous solar processes for lunar integration have been provided and future strategies for solar reactor developments have been proposed. Vacuum pyrolysis offers a promising approach to extract and utilize resources on the Moon, promoting self-sufficiency, reducing costs, and supporting sustainable lunar exploration and potential future missions. However, it remains understudied, with many technical aspects poorly understood, which thus lowers the relative TRL of the process in the minds of many mission planners and technologists.

CRedit authorship contribution statement

Jack Robinot: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sylvain Rodat:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Stéphane Abanades:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Alexis Paillet:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Aidan Cowley:** Supervision, Resources, Project administration, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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